

Big Data Analysis of Competitive Multiplayer Ranking Systems

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1 Introduction

Online competitive environments generate large volumes of data describing player activity, performance, and progression. The increasing availability of these data provides an opportunity to apply Big Data and Data Mining techniques in order to better understand competitive ecosystems and ranking dynamics.

This project focuses on the analysis of two independent competitive multiplayer communities that employ rating-based matchmaking systems. The study investigates how player performance relates to ranking progression and whether meaningful behavioral patterns can be identified within the player population.

The analysis follows the CRISP-DM (Cross Industry Standard Process for Data Mining) methodology, covering data understanding, preparation, modeling, evaluation, and interpretation of the obtained results.

2 CRISP-DM Methodology

The project follows the CRISP-DM framework, a widely adopted methodology for structuring Data Mining projects.

The workflow is divided into five phases:

- **Business Understanding**
- **Data Understanding**
- **Data Preparation**
- **Modeling**
- **Evaluation**

The following sections describe how each phase is applied within the context of this study.

2.1 Business Understanding

The primary objective of this project is to investigate the behavior of competitive ranking systems using real-world player data.

In particular, the analysis aims to answer the following research questions:

- Which performance indicators are most strongly associated with player rating?
- Are there measurable characteristics that distinguish highly ranked players from the rest of the population?
- Can groups of players with similar behavioral profiles be identified?
- How do player ratings evolve over time?
- Are similar patterns observable across different competitive communities?

Answering these questions may provide insights into both player behavior and the effectiveness of ranking-based competitive systems.

2.2 Data Understanding

The study uses data collected from two independent competitive communities, referred to as **Soccer** and **BlockBall**.

After preprocessing, the available data consists of:

- **107** ranked Soccer players;
- **97** ranked BlockBall players;
- more than **34,000** historical rating observations.

The available information can be grouped into three main categories:

- **Player Statistics:** cumulative performance indicators such as goals, assists, saves, wins, losses, draws, and matches played.
- **Ranking Information:** current rating values and ranking-related metrics.
- **Historical Rating Data:** temporal records describing rating evolution over time.

The resulting datasets support two complementary forms of analysis:

- **Cross-sectional analysis**, which compares players at a specific point in time.
- **Longitudinal analysis**, which investigates rating progression throughout a player’s competitive activity.

Before proceeding with modeling, descriptive statistics and exploratory visualizations will be used to evaluate data quality, identify potential anomalies, and understand the overall structure of the datasets.

2.3 Data Preparation

Before performing any analysis, the raw data underwent a preprocessing phase designed to improve consistency, reliability, and privacy.

Anonymization

All player identifiers were anonymized by replacing original UUIDs with artificial identifiers (e.g., Player_001, Player_002, etc.). This procedure preserves privacy while maintaining the possibility of tracking player activity across datasets.

Data Cleaning

Inactive players with no ranked activity were removed from the analysis. Administrative fields unrelated to competitive performance were discarded, and missing values were handled according to the semantic meaning of each variable.

Additionally, anomalous rating values were identified and validated through consistency checks against the available rating history.

Data Transformation

The Soccer dataset originally stored player statistics in a long format, where each row contained a statistic type and its corresponding value. A pivot transformation was therefore applied to obtain a player-level representation suitable for statistical analysis.

The BlockBall dataset was already available in this format and consequently required no equivalent transformation.

Timestamp information was converted into a standard datetime representation, and historical rating records were organized chronologically for each player. A relative temporal variable was also computed, representing the number of days elapsed since each player's first recorded rating observation.

Feature Engineering

To enable fair comparisons between players with different activity levels, several normalized performance indicators were introduced:

- **Win Rate**
- **Goals per Match**
- **Assists per Match**
- **Saves per Match**
- **Passes per Match** (BlockBall only)

Output Datasets

The preprocessing phase produced four cleaned datasets:

- Soccer Player Dataset
- Soccer Rating History Dataset
- BlockBall Player Dataset
- BlockBall Rating History Dataset

These datasets provide a consistent and reproducible representation of player performance and rating progression, suitable for statistical analysis, dimensionality reduction, clustering, and temporal modeling.

2.4 Modeling

The analytical phase employs a combination of statistical and machine learning techniques.

First, descriptive statistics are used to summarize the main characteristics of the datasets and identify relevant distributional properties.

Correlation analysis is then performed to investigate relationships between performance indicators and player ratings.

To explore the underlying structure of the data, **Principal Component Analysis (PCA)** is applied as a dimensionality reduction technique. Clustering algorithms are subsequently used to identify groups of players with similar statistical profiles.

Finally, temporal analyses are conducted on the rating-history datasets in order to investigate ranking progression and rating stability over time.

2.5 Evaluation

The quality and usefulness of the obtained results are assessed through both quantitative and qualitative criteria.

For correlation analysis, the strength and direction of the observed relationships are evaluated. For dimensionality reduction, the proportion of explained variance is considered. Clustering results are assessed according to their interpretability and their ability to identify meaningful player groups.

The temporal analyses are evaluated according to their ability to describe rating evolution and highlight differences between player trajectories.

The final evaluation focuses on determining whether the identified patterns provide meaningful insights into competitive behavior and ranking-system dynamics.

3 Analysis and Results

4 Conclusions